

Juliann Zhu

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EDUCATION

University of Waterloo, GPA: 3.93/4.0

BCS, Honours Computer Science with Co-operative Program

President Scholarship of Distinction

Waterloo, ON

Sep. 2025 - Present

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML, CSS, C, Racket, SQL, Bash

Data and Analytics: MySQL, Grafana, Pandas, NumPy, Matplotlib, InfluxDB

Frameworks and APIs: React, Gemini AI API, Node.js, REST APIs

Tools and Systems: Git, Github, Linux, Google Cloud Platform, Base44, VirtualBox, VS Code

RELEVANT EXPERIENCE

ORBCOMM, SKYWAVE, Network Services Department

May 2026 - Present

Network Engineering Intern

Ottawa, ON

- Developed Grafana dashboards with MySQL data sources to monitor satellite systems, analyze network usage, track service channel performance, and display operational metrics as time-series data visualizations
- Optimized SQL queries to improve dashboard performance and reduce loading time by 86%
- Implemented dynamic filtering systems to enhance user-driven data segmentation across large datasets
- Delivered technical demos for enhanced usability, improved network metric accuracy, SQL-based transformations, UI features, visualization improvements, and AI-driven monitoring progression

Usable Security & Privacy Research, Women in Math DRP, University of Waterloo

Jan. 2026 -Present

Undergraduate Research Mentee

Waterloo, ON

- Co-authoring an academic research paper for the Symposium On Usable Privacy and Security (SOUPS) investigating how users seek security and privacy advice from large language models (LLMs)
- Conducting bi-weekly structured literature reviews to implement prior research in usable security
- Analyzing real-world security and privacy prompts sourced from online databases and evaluating LLM-generated advice against expert feedback from security researchers to assess accuracy and risks

PROJECTS

TrulyHer | React, Javascript, HTML, CSS, Base44, VS Code

Sep. 2025

AI Integration & Front-End Developer

Waterloo, ON

- Developed an AI-powered web application with speech and text input to help women in computer science manage imposter syndrome through journaling and mood detection
- Designed an interactive, user-friendly dashboard with UI/UX principles to visualize emotional trends, generate personalized insights, encourage mental health awareness, and empower underrepresented groups

Project Tech Careers Hackathon

May - Jun. 2024

Platform Designer and UI/UX Developer

Ottawa, ON

- Designed a four-stage platform using Javascript, HTML, and CSS to mentor and build a collection of resources dedicated to supporting women at different stages of their education
- Awarded the *Gender Equality Track Award* for outstanding innovation, design, and impact in addressing the UN Sustainable Development Goals by addressing the gender gap in computer science industries

NeuraLearn | Python, HTML, CSS, Gemini AI, NLP, Markdown, API Integration

Sep. 2024

Front-End Developer & UI/UX Designer

Ottawa, ON

- Built an AI-driven study tool that generates adaptive quizzes and instant Q&A responses based on notes
- Integrated Gemini AI's NLP model to summarize and deliver personalized, data-driven learning experiences
- Engineered custom parsers to convert API data outputs into markdown-formatted quizzes and explanations
- Designed a responsive, intuitive UI for students, emphasizing clarity, accessibility, and reduced cognitive load